Songwei Ge

(+1) 412-320-6392 | songweige.sg@gmail.com | https://songweige.github.io/ | SongweiGe | songwei-ge-898996128

Research Interest

Image, Video, and 3D Synthesis

Education

Department of Computer Science, University of Maryland — *College Park, MD, USA*

Ph.D. in Computer Science — *Sep. 2020 - Present*

- Advisors: David Jacobs and Jia-Bin Huang.
- GPA: 4.00/4.00
- Selected courses: Advanced Numerical Optimization, Fundamental Concepts of Differential Geometry, Foundation of Deep Learning.

School of Computer Science, Carnegie Mellon University — *Pittsburgh, PA, USA*

M.S. in Computational Biology — *Aug. 2018 - May. 2020*

- Thesis Advisors: Barnabás Póczos, Eunsu Kang and Ziv Bar-Joseph.
- GPA: 4.00/4.00
- Selected courses: Machine Learning for Ph.D., Convex Optimization, Probabilistic Graphical Models.

School of Information, Renmin University of China — *Beijing, China*

B.S. in Computer Science and Technology — *Sep. 2014 – Jul. 2018*

- Thesis Advisors: Zhicheng Dou. (**Best thesis award** in Computer Science.)

HillHouse Academy, Renmin University of China — *Beijing, China*

B.S. in Finance — *Sep. 2014 – Jul. 2018*

- Thesis Advisor: Wei Xu.

Employment

NVIDIA Research — *Santa Clara, CA, USA*

Mentors: Yogesh Balaji, Ming-Yu Liu — *May. 2024 - Now*

Research Intern

BAIR, University of California, Berkeley. — *Berkeley, CA, USA*

Mentor: Angjoo Kanazawa — *Aug. 2023 - May. 2024*

Visiting Scholar

The Robotics Institute, Carnegie Mellon University. — *Pittsburgh, PA, USA*

Mentor: Jun-Yan Zhu — *Jan. 2023 - Aug. 2023*

Visiting PhD

NVIDIA Research — *Santa Clara, CA, USA*

Mentors: Yogesh Balaji, Ming-Yu Liu — *May. 2022 - May. 2023*

Research Intern

Facebook AI Research — *Menlo Park, CA, USA*

Mentor: Devi Parikh — *June. 2021 – Dec. 2021*

Research Intern, Student Researcher

Facebook AI Research — *Menlo Park, CA, USA*

Mentors: Larry Zitnick, Devi Parikh — *June. 2020 – Aug. 2020*

Research Intern

Microsoft Research — *Beijing, China*

Advisor: Ruihua Song — *Apr. 2018 – Jul. 2018*

Software Engineer Intern

Selected Publications

Peer Reviewed Publications

- **On the Content Bias in Frechet Video Distance.**
  Songwei Ge, Aniruddha Mahapatra, Gaurav Parmar, Jun-Yan Zhu, Jia-Bin Huang.
  *In Computer Vision and Pattern Recognition (**CVPR**), 2024*
- **Grounded Text-to-Image Synthesis with Attention Refocusing.**
  Quynh Phung, Songwei Ge, Jia-Bin Huang.
  *In Computer Vision and Pattern Recognition (**CVPR**), 2024*
- **Preserve Your Own Correlation: A Noise Prior for Video Diffusion Models.**
  Songwei Ge, Guilin Liu, Seungjun Nah, Tyler Poon, Andrew Tao, Bryan Catanzaro, David Jacobs, Jia-Bin Huang, Ming-Yu Liu, Yogesh Balaji.
  *In International Conference on Computer Vision **(ICCV)**, 2023*
- **Expressive Text-to-Image Generation with Rich Text.**
  Songwei Ge, Taesung Park, Jun-Yan Zhu, Jia-Bin Huang.
  *In International Conference on Computer Vision **(ICCV)**, 2023*
- **Hyperbolic Contrastive Learning for Visual Representations beyond Objects.**
  Songwei Ge*, Shlok Mishra*, Simon Kornblith, Chun-Liang Li, David Jacobs.
  *In Computer Vision and Pattern Recognition (**CVPR**), 2023*
- **Long Video Generation with Time-Agnostic VQGAN and Time-Sensitive Transformer.**
  Songwei Ge, Thomas Hayes, Harry Yang, Xi Yin, Guan Pang, David Jacobs, Jia-Bin Huang, Devi Parikh.
  *In European Conference on Computer Vision (**ECCV**), 2022.*
- **MUGEN: A Playground for Video-Audio-Text Multimodal Understanding and GENeration.**
  Thomas Hayes, Songyang Zhang, Xi Yin, Guan Pang, Sasha Sheng, Harry Yang, Songwei Ge, Isabelle Hu, Devi Parikh
  *In European Conference on Computer Vision (**ECCV**), 2022.*
- **Robust Contrastive Learning Using Negative Samples with Diminished Semantics.**
  Songwei Ge, Shlok Mishra, Haohan Wang, Chun-Liang Li, David Jacobs.
  *In Neural Information Processing Systems (**NeurIPS**), 2021.*
- **Shift Invariance Can Reduce Adversarial Robustness.**
  Songwei Ge*, Vasu Singla*, Ronen Basri, David Jacobs.
  *In Neural Information Processing Systems (**NeurIPS**), 2021.*
- **Conceptual Blending with Large-scale Language and Vision Models.**
  Songwei Ge, Devi Parikh.
  *In International Conference on Computational Creativity (**ICCC Oral**), 2021.*
- **Creative Sketch Generation.**
  Songwei Ge, Vedanuj Goswami, Larry Zitnick, Devi Parikh.
  *In Workshop on Machine Learning for Creativity and Design at NeurIPS (**Oral**), 2020*
  *In International Conference on Learning Representations (**ICLR**) 2021.*
- **Supervised Adversarial Alignment of Single-Cell RNA-seq Data.**
  Songwei Ge, Haohan Wang, Amir Alavi, Eric P. Xing, Ziv Bar-Joseph.
  *In the 24th International Conference on Research in Computational Molecular Biology (**RECOMB**), 2020*
- **Learning Robust Global Representations by Penalizing Local Predictive Power.**
  Haohan Wang, Songwei Ge, Eric P. Xing, Zachary C. Lipton.
  *In Neural Information Processing Systems (**NeurIPS**), 2019.*
- **From Text to Sound: a Preliminary Study on Retrieving Sound Effects to Radio Stories.**
  Songwei Ge, Curtis Xuan, Ruihua Song, Chao Zou, Wei Liu and Jin Zhou.
  *In ACM International Conference on Research and Development in Information Retrieval (**SIGIR**), 2019.*
- **Hallucinating Point Cloud into 3D Sculptural Object.**
  Songwei Ge*, Chun-Liang Li*, Eunsu Kang*, Lingyao Zhang, Austin Dill, Manzil Zaheer, Barnabás Póczos.
  *In Workshop on Machine Learning for Creativity and Design at NeurIPS, 2018, and Inter-Society for the Electronic Arts (**ISEA**), 2019*
- **A Computational Strategy for Finding Novel Targets and Therapeutic Compounds for Opioid Dependence.**
  Xiaojun Wu, Siwei Xie, Lirong Wang, Peihao Fan, Songwei Ge, Xiang-Qun Xie, Wei Wu.

In PLoS One, 2018

- **Personalizing Search Results Using Hierarchical RNN with Query-aware Attention.**
  Songwei Ge, Zhicheng Dou, Zhengbao Jiang, Jian-Yun Nie, Ji-Rong Wen.
  *In ACM International Conference on Information and Knowledge Management (**CIKM**), 2018*

Selected Honors & Awards

Dec 2023	**NVIDIA Graduate Fellowship**	*Santa Clara, CA*
Jan 2023	**Meta Graduate Fellowship Finalist**	*Menlo Park, CA*
Dec 2022	**NVIDIA Graduate Fellowship Finalist**	*Santa Clara, CA*
Dec 2022	**NeurIPS Outstanding Reviewer**	*New Orleans, LA*
2019-2020	**CMU MSCB Academic Scholarship**	*Pittsburgh, PA*
Jun 2018	**Outstanding Undergraduate Thesis Award** (Top 1% in RUC & Top 1 in the Computer Science)	*Beijing, China*
Jan 2017	**Meritorious Winners,** American Mathematical Contest in Mathematical Modeling	*Beijing, China*

Professional Service

2021-2024	**Conference Reviewer** Annual Conference on Neural Information Processing Systems	*NeurIPS*
2023	**Conference Reviewer** ACM SIGGRAPH Conference and Exhibition on Computer Graphics in Asia	*SIGGRAPH Asia*
2022-2024	**Conference Reviewer** International Conference on Learning Representations	*ICLR*
2023	**Conference Reviewer** ACM SIGGRAPH Conference and Exhibition on Computer Graphics	*SIGGRAPH*
2023	**Conference Reviewer** International Conference on Computer Vision	*ICCV*
2023	**Jounal Reviewer** Transactions on Pattern Analysis and Machine Intelligence	*PAMI*
2023-2024	**Conference Reviewer** Conference on Computer Vision and Pattern Recognition	*CVPR*
2022	**External Reviewer** The ACM Symposium on User Interface Software and Technology	*UIST*
2022-2024	**Conference Reviewer** European Conference on Computer Vision	*ECCV*
2022	**Emergency Reviewer** Conference on Computer Vision and Pattern Recognition	*CVPR*
2022-2024	**Conference Reviewer** International Conference on Machine Learning	*ICML*

Teaching

2023	**Guest Lecturer** Art and Machine Learning	*Pittsburgh, PA*
2021	**Teaching Assistant** Algorithm	*College Park, MD*
2020	**Teaching Assistant** Introduction to Machine Learning	*College Park, MD*
2019	**Guest Lecturer** Art and Machine Learning	*Pittsburgh, PA*
2017	**Teaching Assistant** Advanced Algorithm Design	*Beijing China*